SET

In [33]:
```
s = {}   # set & dict both define with {}
s
```

Out[33]: {}

In [34]:
```
type(s)
```

Out[34]: dict

In [35]:
```
s1 = set()
type(s1)
```

Out[35]: set

In [36]:
```
s2 = {20,100,3,45}   # if you send same date type  it will give in (
s2
```

Out[36]: {3, 20, 45, 100}

In [37]:
```
s3 = {'z','l','c','e','f'}  # it will give ordered way
s3
```

Out[37]: {'c', 'e', 'f', 'l', 'z'}

In [38]:
```
s4 = {2,3,4,'nit',1+2j,False}
s4
```

Out[38]: {(1+2j), 2, 3, 4, False, 'nit'}

In [39]:
```
print(s1)
print(s2)
print(s3)
print(s4)
```

```
set()
{45, 3, 100, 20}
{'c', 'z', 'l', 'f', 'e'}
{False, 2, 3, 4, 'nit', (1+2j)}
```

In [40]:
```
s2
```

Out[40]: {3, 20, 45, 100}

In [41]:
```
s2.add(30)
```

In [42]:
```
s2
```

Out[42]: {3, 20, 30, 45, 100}

In [43]:
```
s2.add(200)
```

In [44]:
```
s2
```

Out[44]: {3, 20, 30, 45, 100, 200}

In [45]:
```
s2
```

Out[45]: {3, 20, 30, 45, 100, 200}

In [46]:
```
 #s2[1:3] # in set indexing and slicing will not allowed
```

In [47]:
```
s4
```

Out[47]: {(1+2j), 2, 3, 4, False, 'nit'}

In [48]:
```
s5 = s4.copy()
s5
```

Out[48]: {(1+2j), 2, 3, 4, False, 'nit'}

In [49]:
```
s5.add(2)  # duplicates are not allowed
```

In [50]:
```
s5
```

Out[50]: {(1+2j), 2, 3, 4, False, 'nit'}

In [51]:
```
s4
```

Out[51]: {(1+2j), 2, 3, 4, False, 'nit'}

In [52]:
```
s5.clear()
s5
```

Out[52]: set()

In [53]:
```
del s5
```

In [54]:
```
s4
```

Out[54]: {(1+2j), 2, 3, 4, False, 'nit'}

In [55]:
```
s4 = {(1+2j), 2, 3, 4, False, 'nit'}
s4
```

Out[55]: {(1+2j), 2, 3, 4, False, 'nit'}

In [56]:
```
s4.remove((1+2j)) # it takes only one argument
```

In [57]:
```
s4
```

Out[57]: {2, 3, 4, False, 'nit'}

In [58]:
```
s3
```

Out[58]: {'c', 'e', 'f', 'l', 'z'}

In [59]:
```
s3.discard('m') # in set m in not there but wont through an "error"
```

In [60]:
```
s3.discard('f')
s3
```

Out[60]: {'c', 'e', 'l', 'z'}

In [61]:
```
s3.pop()  # it will delete random in list it delete last index
s3
```

Out[61]: {'e', 'l', 'z'}

In [62]:
```
s2
```

Out[62]: {3, 20, 30, 45, 100, 200}

In [63]:
```
s2.pop(3)  # indexing is not alllowed
s2
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[63], line 1
----> 1 s2.pop(3)  # indexing is not alllowed
      2 s2

TypeError: set.pop() takes no arguments (1 given)
```

In [64]:
```
s2.pop()
s2
```

Out[64]: {20, 30, 45, 100, 200}

In [65]:
```
for i in enumerate(s2):
    print(i)
```

```
(0, 100)
(1, 200)
(2, 45)
(3, 20)
(4, 30)
```

In [66]:
```
for i in s2:
    print(i)
```

```
100
200
45
20
30
```

In [67]:
```
100 in s2
```

Out[67]: True

In [68]:
```
1000 in s2
```

Out[68]: False

In [69]:
```
s2.update(s3)
s2
```

Out[69]: {100, 20, 200, 30, 45, 'e', 'l', 'z'}

SET OPERATION

In [70]:
```
s6 = {1,2,3,4,5}
s7 = {4,5,6,7,8}
s8 = {8,9,10}
```

In [71]:
```
s6.union(s7) # duplication is NOT allowed
```

Out[71]: {1, 2, 3, 4, 5, 6, 7, 8}

In [72]:
```
s6.union(s7,s8)
```

Out[72]: {1, 2, 3, 4, 5, 6, 7, 8, 9, 10}

In [73]:
```
s6 | s7
```

Out[73]: {1, 2, 3, 4, 5, 6, 7, 8}

In [74]:
```
s6 | s7 | s8
```

Out[74]: {1, 2, 3, 4, 5, 6, 7, 8, 9, 10}

In [75]:
```
print(s6)
print(s7)
print(s8)
```

```
{1, 2, 3, 4, 5}
{4, 5, 6, 7, 8}
{8, 9, 10}
```

In [76]:
```
s6.intersection(s7) # common element will display
```

Out[76]: {4, 5}

In [77]:
```
s6.intersection(s8)
```

Out[77]: set()

In [78]:
```
s7.intersection(s8)
```

Out[78]: {8}

In [79]:
```
s6 & s7
```

Out[79]: {4, 5}

In [80]:
```
print(s6)
print(s7)
print(s8)
```

```
{1, 2, 3, 4, 5}
{4, 5, 6, 7, 8}
{8, 9, 10}
```

In [81]:
```
s6.difference(s7) # common will not print
```

Out[81]: {1, 2, 3}

In [82]:
```
s6 - s7
```

Out[82]: {1, 2, 3}

In [83]:
```
s6 - s8
```

Out[83]: {1, 2, 3, 4, 5}

In [84]:
```
s8 - s6
```

Out[84]: {8, 9, 10}

In [85]:
```
s7 - s8
```

Out[85]: {4, 5, 6, 7}

In [86]:
```
s8 - s7
```

Out[86]: {9, 10}

In [87]:
```
print(s6)
print(s7)
print(s8)
```

```
{1, 2, 3, 4, 5}
{4, 5, 6, 7, 8}
{8, 9, 10}
```

In [88]:
```
s6.symmetric_difference(s7) # it will remove common elements and ot
```

Out[88]: {1, 2, 3, 6, 7, 8}

In [89]:
```
s6 ^ s7
```

Out[89]: {1, 2, 3, 6, 7, 8}

In [90]:
```
s11 = {1,2,3,4,5,6,7,8,9}
s12 = {3,4,5,6,7,8}
s13 = {10,20,30,40}
```

In [91]:
```
s12.issubset(s11)
```

Out[91]: True

In [92]:
```
s11.issubset(s12)
```

Out[92]: False

In [93]:
```
s11.issuperset(s12)
```

Out[93]: True

In [94]:
```
s11 = {1,2,3,4,5,6,7,8,9}
s12 = {3,4,5,6,7,8}
s13 = {10,20,30,40}
```

In [95]:
```
s13.isdisjoint(s12)
```

Out[95]: True

In [96]:
```
s13.isdisjoint(s11)
```

Out[96]: True

In [97]:
```
s12.isdisjoint(s11)
```

Out[97]: False

In [98]:
```
s12 = {1,2,3,4,5}
s13 = {10,20,30}
s14 = {15,25,35}
```

In [99]:
```
s13.issubset(s12)
```

Out[99]: False

In [100…
```
s12.issuperset(s13)
```

Out[100… False

In [101…
```
 s11
```

Out[101… {1, 2, 3, 4, 5, 6, 7, 8, 9}

In [102…
```
for i in s11:
    print(i)
```

```
1
2
3
4
5
6
7
8
9
```

In [103…
```
for i in enumerate(s11):
    print(i)
```

```
(0, 1)
(1, 2)
(2, 3)
(3, 4)
(4, 5)
(5, 6)
(6, 7)
(7, 8)
(8, 9)
```

In [104…
```
sum(s11)
```

Out[104… 45

In [105…
```
min(s11)
```

Out[105… 1

In [106…
```
max(s11)
```

Out[106… 9

In [107…
```
list(enumerate(s11))
```

Out[107… [(0, 1), (1, 2), (2, 3), (3, 4), (4, 5), (5, 6), (6, 7), (7, 8), (8, 9)]

In [108…
```
sorted(s11)
```

Out[108… [1, 2, 3, 4, 5, 6, 7, 8, 9]

Dictionary

In [109…
```
d = {}
```

In [110…
```
d
```
Out[110… {}

In [111…
```
type(d)
```
Out[111… dict

In [112…
```
d1 = {1:'one',2:'two',3:'three'}
d1
```
Out[112… {1: 'one', 2: 'two', 3: 'three'}

In [113…
```
d1.keys()
```
Out[113… dict_keys([1, 2, 3])

In [114…
```
d1.values()
```
Out[114… dict_values(['one', 'two', 'three'])

In [115…
```
d2 = d1.copy()
d2
```
Out[115… {1: 'one', 2: 'two', 3: 'three'}

In [116…
```
d1.items()
```
Out[116… dict_items([(1, 'one'), (2, 'two'), (3, 'three')])

In [117…
```
d1[1]
```
Out[117… 'one'

In [118…
```
keys = {'ram','b','c','d'}
value = [10,20,30]
mydict3 = dict.fromkeys(keys,value)
mydict3
```
Out[118… {'ram': [10, 20, 30], 'c': [10, 20, 30], 'd': [10, 20, 30], 'b': [10, 20, 30]}

In [119…
```
value.append(50)
```
In [120…
```
mydict3
```
Out[120…
```
{'ram': [10, 20, 30, 50],
 'c': [10, 20, 30, 50],
 'd': [10, 20, 30, 50],
 'b': [10, 20, 30, 50]}
```
In [121…
```
range(10)
```

Out[121… range(0, 10)

In [122…
```
list(range(0,10))
```

Out[122… [0, 1, 2, 3, 4, 5, 6, 7, 8, 9]

In [123…
```
r = range(0,10)
r
```

Out[123… range(0, 10)

In [124…
```
for i in r:
    print(i)
```

```
0
1
2
3
4
5
6
7
8
9
```

In [125…
```
for i in enumerate(r):
    print(i)
```

```
(0, 0)
(1, 1)
(2, 2)
(3, 3)
(4, 4)
(5, 5)
(6, 6)
(7, 7)
(8, 8)
(9, 9)
```

Set creation

In [126…
```
myset = {1,2,3,4,5} # Set of numbers
myset
```

Out[126… {1, 2, 3, 4, 5}

In [127…
```
len(myset)
```

Out[127… 5

In [128…
```
my_set = {1,1,2,2,3,4,5,5}  # Duplicate elements are not allowed
my_set
```

Out[128… {1, 2, 3, 4, 5}

In [129…
```
myset1 = {1.79,2.08,3.99,4.56,5.45} # Set of float numbers
myset1
```

Out[129… {1.79, 2.08, 3.99, 4.56, 5.45}

In [130…
```
myset2 = {'abhi','rami','reddy'}
myset2
```

Out[130… {'abhi', 'rami', 'reddy'}

In [131…
```
myset3 = {10,20,(1,2,3),'abhi',1.2}
myset3
```

Out[131… {(1, 2, 3), 1.2, 10, 20, 'abhi'}

In [132…
```
myset3 = {10,20, "Hola", [11, 22, 32]} # set doesn't allow mutable 
myset3
```

```
---------------------------------------------------------------------
-------
TypeError                                 Traceback (most recent cal
l last)
Cell In[132], line 1
----> 1 myset3 = {10,20, "Hola", [11, 22, 32]} # set doesn't allow m
utable items like li
      2 myset3

TypeError: unhashable type: 'list'
```

In [133…
```
myset4 = set()  # Creation empty set
type(myset4)
```

Out[133… set

In [134…
```
my_set1 = set(('one' , 'two' , 'three' , 'four'))
my_set1
```

Out[134… {'four', 'one', 'three', 'two'}

Loop through set

In [135…
```
myset ={'one','two','three','four','five','six','seven','eight'}
myset
```

Out[135… {'eight', 'five', 'four', 'one', 'seven', 'six', 'three', 'two'}

In [136…
```
myset
```

Out[136… {'eight', 'five', 'four', 'one', 'seven', 'six', 'three', 'two'}

In [137…
```
for i in myset:
    print(i)
```

```
six
seven
four
one
eight
two
five
three
```

In [138…
```
for i in enumerate(myset):
    print(i)
```

```
(0, 'six')
(1, 'seven')
(2, 'four')
(3, 'one')
(4, 'eight')
(5, 'two')
(6, 'five')
(7, 'three')
```

Set memebership

In [139…
```
myset
```

Out[139…
```
{'eight', 'five', 'four', 'one', 'seven', 'six', 'three', 'two'}
```

In [140…
```
'one' in myset
```

Out[140…
```
True
```

In [141…
```
'nine' in myset
```

Out[141…
```
False
```

In [142…
```
if 'three' in myset:
    print('Three is present in set')
else:
    print('Three is not present in set')
```

```
Three is present in set
```

Add and Remove

In [143…
```
myset
```

Out[143…
```
{'eight', 'five', 'four', 'one', 'seven', 'six', 'three', 'two'}
```

In [144…
```
myset.add('nine')
myset
```

Out[144… {'eight', 'five', 'four', 'nine', 'one', 'seven', 'six', 'three', 'two'}

In [145…
```
myset.remove('nine')
myset
```

Out[145… {'eight', 'five', 'four', 'one', 'seven', 'six', 'three', 'two'}

In [146…
```
myset.add('nine')
myset
```

Out[146… {'eight', 'five', 'four', 'nine', 'one', 'seven', 'six', 'three', 'two'}

In [147…
```
myset.discard('nine')
```

In [148…
```
myset
```

Out[148… {'eight', 'five', 'four', 'one', 'seven', 'six', 'three', 'two'}

In [149…
```
myset.update('nine','ten','eleven')
myset
```

Out[149…
```
{'e',
 'eight',
 'five',
 'four',
 'i',
 'l',
 'n',
 'one',
 'seven',
 'six',
 't',
 'three',
 'two',
 'v'}
```

In [150…
```
del myset
```

Copy Set

In [151…
```
myset = {'one', 'two', 'three', 'four', 'five', 'six', 'seven', 'ei
myset
```

Out[151… {'eight', 'five', 'four', 'one', 'seven', 'six', 'three', 'two'}

In [152…
```
myset1 = myset
myset1
```

Out[152… {'eight', 'five', 'four', 'one', 'seven', 'six', 'three', 'two'}

In [153…
```
id(myset) , id(myset1) # The address of both myset & myset1 will be
```

Out[153… (4924829568, 4924829568)

In [154…
```
my_set = myset.copy()
```

In [155…
```
my_set
```

Out[155… {'eight', 'five', 'four', 'one', 'seven', 'six', 'three', 'two'}

In [156…
```
id(my_set),id(myset) # Here we can get different address
```

Out[156… (4924829792, 4924829568)

Set operations

In [157…
```
A = {1,2,3,4,5}
B = {4,5,6,7,8}
C = {8,9,10}
```

In [158…
```
A | B,C
```

Out[158… ({1, 2, 3, 4, 5, 6, 7, 8}, {8, 9, 10})

In [159…
```
A.union(B)
```

Out[159… {1, 2, 3, 4, 5, 6, 7, 8}

In [160…
```
A.union(B,C)
```

Out[160… {1, 2, 3, 4, 5, 6, 7, 8, 9, 10}

In [161…
```
A.intersection(B)
```

Out[161… {4, 5}

In [162…
```
A & B
```

Out[162… {4, 5}

In [163…
```
A & C
```

Out[163… set()

In [164…
```
A.difference(B)
```

Out[164… {1, 2, 3}

In [165…
```
A -B
```

Out[165… {1, 2, 3}

In [166…
```
A - C
```
Out[166… {1, 2, 3, 4, 5}

In [167…
```
A.symmetric_difference(B)
```
Out[167… {1, 2, 3, 6, 7, 8}

In [168…
```
A.issubset(B)
```
Out[168… False

In [169…
```
A.issuperset(B)
```
Out[169… False

In [170…
```
A.isdisjoint(B)
```
Out[170… False

Other built in functions

In [171…
```
A
```
Out[171… {1, 2, 3, 4, 5}

In [172…
```
min(A)
```
Out[172… 1

In [173…
```
max(A)
```
Out[173… 5

In [174…
```
list(enumerate(A))
```
Out[174… [(0, 1), (1, 2), (2, 3), (3, 4), (4, 5)]

In [175…
```
D = sorted(A,reverse = True)
D
```
Out[175… [5, 4, 3, 2, 1]

In [176…
```
D = sorted(A,reverse = False)
D
```
Out[176… [1, 2, 3, 4, 5]

Dictionary

Dictionary is a mutable data type in Python. A python dictionary is a collection of key and value pairs separated by a colon (:) & enclosed in curly braces {}. Keys must be unique in a dictionary, duplicate values are allowed.

In [177…
```
mydict = dict() # empty dictionary
mydict
```

Out[177…
```
{}
```

In [178…
```
mydict = {1:'one' , 2:'two' , 3:'three'} # dictionary with integer
mydict
```

Out[178…
```
{1: 'one', 2: 'two', 3: 'three'}
```

In [179…
```
mydict = dict({1:'one' , 2:'two' , 3:'three'}) # Create dictionary
mydict
```

Out[179…
```
{1: 'one', 2: 'two', 3: 'three'}
```

In [180…
```
mydict = {'A':'one' , 'B':'two' , 'C':'three'} # dictionary with ch
mydict
```

Out[180…
```
{'A': 'one', 'B': 'two', 'C': 'three'}
```

In [182…
```
mydict = {1:'one' , 'A':'two' , 3:'three'} # dictionary with mixed
mydict
```

Out[182…
```
{1: 'one', 'A': 'two', 3: 'three'}
```

In [183…
```
mydict.keys()
```

Out[183…
```
dict_keys([1, 'A', 3])
```

In [184…
```
mydict.values()
```

Out[184…
```
dict_values(['one', 'two', 'three'])
```

In [185…
```
mydict.items()
```

Out[185…
```
dict_items([(1, 'one'), ('A', 'two'), (3, 'three')])
```

In [187…
```
mydict = {1:'one' , 2:'two' , 'A':['asif' , 'john' , 'Maria']} # di
mydict
```

Out[187…
```
{1: 'one', 2: 'two', 'A': ['asif', 'john', 'Maria']}
```

In [188…
```
keys = {'a' , 'b' , 'c' , 'd'}
value = 10
mydict3
mydict3 = dict.fromkeys(keys , value) # Create a dictionary from a
```

In [189…
```
mydict3
```

Out[189… {'c': 10, 'd': 10, 'b': 10, 'a': 10}

In [194…
```
keys = {'a' , 'b' , 'c' , 'd'}
value = [10,20,30]
mydict3 = dict.fromkeys(keys , value) # Create a dictionary from a
mydict3
```

Out[194… {'c': [10, 20, 30], 'd': [10, 20, 30], 'b': [10, 20, 30], 'a': [1
0, 20, 30]}

In [195…
```
value.append(40)
```

In [196…
```
mydict3
```

Out[196…
```
{'c': [10, 20, 30, 40],
 'd': [10, 20, 30, 40],
 'b': [10, 20, 30, 40],
 'a': [10, 20, 30, 40]}
```

Accessing Items

In [197…
```
mydict = {1:'one' , 2:'two' , 3:'three' , 4:'four'}
mydict
```

Out[197… {1: 'one', 2: 'two', 3: 'three', 4: 'four'}

In [198…
```
mydict[1]
```

Out[198… 'one'

In [199…
```
mydict.get(1)
```

Out[199… 'one'

In [200…
```
mydict1 = {'Name':'Asif' , 'ID': 74123 , 'DOB': 1991 , 'job' :'Anal
mydict1
```

Out[200… {'Name': 'Asif', 'ID': 74123, 'DOB': 1991, 'job': 'Analyst'}

In [201…
```
mydict1['Name']
```

Out[201… 'Asif'

In [202…
```
mydict1['DOB']
```

Out[202… 1991

Change and accesssing

In [203…
```
mydict1 = {'Name':'Asif' , 'ID': 12345 , 'DOB': 1991 , 'Address' :
mydict1
```

Out[203… {'Name': 'Asif', 'ID': 12345, 'DOB': 1991, 'Address': 'Hilsinki'}

In [204…
```
mydict1['DOB'] = 2005
mydict1
```

Out[204… {'Name': 'Asif', 'ID': 12345, 'DOB': 2005, 'Address': 'Hilsinki'}

In [206…
```
dict1 = {'DOB':1995}
dict1.update()
dict1
```

Out[206… {'DOB': 1995}

In [207…
```
mydict1['Job'] = 'Analyst' # Adding items in the dictionary
mydict1
```

Out[207…
```
{'Name': 'Asif',
 'ID': 12345,
 'DOB': 2005,
 'Address': 'Hilsinki',
 'Job': 'Analyst'}
```

Loop through dict

In [208…
```
mydict1
```

Out[208…
```
{'Name': 'Asif',
 'ID': 12345,
 'DOB': 2005,
 'Address': 'Hilsinki',
 'Job': 'Analyst'}
```

In [211…
```
for i in mydict1:
    print(i,':',mydict1[i])
```

```
Name : Asif
ID : 12345
DOB : 2005
Address : Hilsinki
Job : Analyst
```

In [ ]: